Why PA$^-$ cannot prove that this machine runs forever

The translated cycler `1RB1RA_1LA---`

The claim. This machine really does run forever from a blank tape. Nevertheless, PA$^-$ cannot prove its nonhalting sentence for the project's binary-stack, quotient-and-remainder history encoding. We establish this by constructing an alternative arithmetic setting that satisfies every PA$^-$ axiom but accepts an apparent halting history.

This is a mathematical proof with an explicit construction, not a Metamath-verified independence theorem. It concerns the precise encoding on pages 2–3, not every possible formulation of nonhalting. No knowledge of models or nonstandard numbers is assumed. Pages 1–6 introduce the idea; pages 7–8 supply the essential coding proof; pages 9–10 explain the logical conclusion and give the axioms and references.

1. The real machine never halts

The tape extends in both directions and contains only 0s initially. A head starts at position 0 in state A. Each instruction reads one bit, writes a bit, moves one cell, and changes state.

State	Read	Instruction
A	0	Write 1, move right, enter B
A	1	Write 1, move right, remain in A
B	0	Write 1, move left, enter A
B	1	Halt

The name `1RB1RA_1LA---` records these four cases in that order. The only way to halt is to be in state B while reading 1.

After $4k$ steps, the head is at cell $2k$ in state A. Cells $0, \ldots, 2k-1$ contain 1; everything else contains 0. The next four steps have this form:

Time	State	Head	Cells containing 1
$4k$	A	$2k$	$0, \ldots, 2k-1$
$4k+1$	B	$2k+1$	$0, \ldots, 2k$
$4k+2$	A	$2k$	$0, \ldots, 2k+1$
$4k+3$	A	$2k+1$	$0, \ldots, 2k+1$
$4k+4$	A	$2k+2$	$0, \ldots, 2k+1$

For $k = 0$, the first interval is empty. The four instructions are $A0$, $B0$, $A1$, $A1$. The final row is the starting pattern shifted two cells right. In particular, every actual visit to B reads a fresh 0. Ordinary induction proves nonhalting.

That last sentence uses induction in our mathematical reasoning. It does not automatically give a proof from the weaker axioms of PA$^-$. To show that no such proof exists, we must do more than notice that this particular argument uses induction.

2. The fixed arithmetic description of a computation

Arithmetic has no primitive tapes or lists. The existing encoding describes one configuration by four numbers

$$c = (q, L, b, R).$$

Here q is the state, b the scanned bit, and L, R are the tape on either side of the head. The nearest cell is the least significant binary digit. Thus the nearest-first list $1, 0, 1, 0, \ldots$ has stack value 5, since $5 = 1 + 0 \cdot 2 + 1 \cdot 4$. Popping it gives bit 1 and remainder stack 2, expressed by $5 = 2 \cdot 2 + 1$.

We retain the project's labels $A = 0$, $B = 1$, $Z = 3$, and pad its three-state interface with an unused state $C = 2$. Represent the missing $B1$ instruction by `1LZ`: write 1, move left, and enter Z. Give both unused C entries the same halting instruction. No A or B instruction enters C. This fixes the recorded halting convention without altering the ordinary blank-tape behavior.

A right-moving instruction writing 1 requires

$$L' = 2L + 1, \qquad R = 2R' + b', \qquad b' \in \{0, 1\}.$$

A left-moving instruction writing 1 requires

$$R' = 2R + 1, \qquad L = 2L' + b', \qquad b' \in \{0, 1\}.$$

The next state q' is specified by the instruction. The fixed-table formula $\mathsf{Step}(c, c')$ is the disjunction of these cases (the last row abbreviates two branches):

Current (q, b)	Next q'	Stack equations	New bit
$(0, 0)$	1	$L' = 2L + 1,\ R = 2R' + b'$	$b' = 0 \vee b' = 1$
$(0, 1)$	0	$L' = 2L + 1,\ R = 2R' + b'$	$b' = 0 \vee b' = 1$
$(1, 0)$	0	$R' = 2R + 1,\ L = 2L' + b'$	$b' = 0 \vee b' = 1$
$(1, 1)$	3	$R' = 2R + 1,\ L = 2L' + b'$	$b' = 0 \vee b' = 1$
$(2, 0)$ or $(2, 1)$	3	$R' = 2R + 1,\ L = 2L' + b'$	$b' = 0 \vee b' = 1$

The project's instruction coding $4q' + 2\delta + w$ gives the table $(7, 3, 1, 13, 13, 13)$. There are no outgoing transitions from state 3. The constructed record, like the genuine run, never uses state 2.

One number for each column of the record

Fix a proposed time n, a bound H, and eight numbers $B_0, C_0, \ldots, B_3, C_3$. Column j is read using the changing divisor

$$D_j(i) = 1 + (i + 1)C_j, \qquad \beta_j(i, v) \ \equiv\ \exists u < H\,[B_j = uD_j(i) + v \ \wedge\ v < D_j(i)].$$

This says that v is a remainder of B_j, with a quotient witness below H. The columns, in order, are q, L, b, R. Put

$$\mathsf{At}(i, (q, L, b, R)) \ \equiv\ \beta_0(i, q) \wedge \beta_1(i, L) \wedge \beta_2(i, b) \wedge \beta_3(i, R).$$

All the displayed names abbreviate formulas using only $0, 1, +, \cdot, <, =$ and logical symbols. Exponentiation is not added to the language. This is the existing history format [4, §§3–4].

3. The statement to be proved unprovable

A tuple $c < H$ means that all four of its entries are below H. Quantifiers over tuples abbreviate four ordinary number quantifiers. The history-checking formula is precisely the conjunction

$$\begin{aligned}
\mathsf{Run} \;\equiv\;\; & \forall i < n+1\ \exists c < H\ \mathsf{At}(i,c), && \text{a row at every time;}\\
& \forall c < H\,[\mathsf{At}(0,c) \to c = (0,0,0,0)], && \text{the blank start;}\\
& \forall i < n\ \forall c,e < H\,[(\mathsf{At}(i,c) \wedge \mathsf{At}(i+1,e)) \to \mathsf{Step}(c,e)], && \text{legal adjacent rows;}\\
& \exists c < H\,[\mathsf{At}(n,c) \wedge q(c) = 3]. && \text{a final halt.}
\end{aligned}$$

Define the two arithmetic sentences

$$\mathsf{Halt}_M \;\equiv\; \exists n, H, B_0, C_0, \ldots, B_3, C_3\ \mathsf{Run}, \qquad \mathsf{Nonhalt}_M \;\equiv\; \neg\mathsf{Halt}_M.$$

The target is PA$^-$ $\nvdash$ $\mathsf{Nonhalt}_M$: the symbol $\nvdash$ means "does not have a formal proof of." Notice that the definition checks *all* times and *all* decoded rows in the proposed interval. A counterexample must meet those conditions, including the existence of a single numerical code for each column.

What is a model?

A formal language supplies symbols and rules; an **interpretation** gives those symbols meanings and specifies what objects the variables range over. A **model of a theory** is an interpretation in which every axiom is true.

The ordinary natural numbers, with their usual operations, are one model of PA$^-$. But a short list of axioms need not characterize only that familiar setting. Another collection of objects might also satisfy all those axioms, even if its objects do not all behave like ordinary finite numbers.

The logical bridge is the **soundness theorem**: a formal proof from given axioms has a true conclusion in *every* model of those axioms. [2] Therefore, to rule out a PA$^-$ proof of $\mathsf{Nonhalt}_M$, it is enough to build

$$\mathcal{M} \models \text{PA}^- \qquad \text{and} \qquad \mathcal{M} \models \mathsf{Halt}_M.$$

The notation $\models$ means "is true in that interpretation." Such a setting is called a **countermodel** to the proposed conclusion $\mathsf{Nonhalt}_M$.

What this requires, in plain language

We will build an integer-like arithmetic setting in which the formula above accepts a halting record. The record will agree with the real run at every ordinary time $0, 1, 2, \ldots$. It will also have a separate terminal part ending in state 3. Its time interval contains both parts but no adjacent pair connecting them. The checks of adjacent times will consequently see no illegal jump.

An ordinary finite interval cannot have this shape. The next page explains, concretely, how an interval in another model can.

4. A nonstandard number, introduced without model theory

Think of polynomials such as $X - 7$, $2X + 3$, and X^2 as *objects*, not as formulas waiting for an ordinary numerical value of X. Add and multiply them by the usual polynomial rules.

Order them by treating X as larger than every ordinary integer. Precisely: a nonzero polynomial is positive when its highest nonzero coefficient is positive; $f < g$ means $g - f$ is positive. Then, for example,

$$100 < X - 7 < X - 1 < X < X + 1 < 2X < X^2.$$

We do not substitute an enormous finite integer for X. The comparison rule defines a new ordered arithmetic setting.

Keeping negative values during the construction

A **ring** is a collection closed under addition, subtraction, and multiplication, with the familiar algebraic laws. The integer-coefficient polynomials form the ring $\mathbb{Z}[X]$. After constructing a suitable ordered ring, we keep just its nonnegative elements as the domain of arithmetic. Negative values are tools for building the setting; they are not in that arithmetic domain.

There is no polynomial strictly between 0 and 1 in this order. A positive nonconstant polynomial exceeds every ordinary integer, while a positive constant is an ordinary positive integer. In an ordered ring with this property, called **discrete order**, the nonnegative part satisfies the PA$^-$ axioms: the algebraic laws come from the ring, and discreteness supplies the successor/order law. Appendix A checks this explicitly. [1]

The surprising shape of a bounded interval

Among nonnegative integer-coefficient polynomials, the interval from 0 to X consists exactly of

$$I = \{0, 1, 2, \ldots\} \cup \{X - r : r = 0, 1, 2, \ldots\}.$$

To verify this, a polynomial of degree at least 2 cannot lie in the interval. A degree-one member must have coefficient 1 on X, so it is $X - r$. The remaining members are ordinary nonnegative constant polynomials.

Part of the interval	Typical times	Its endpoint behavior
Initial part	$0, 1, 2, \ldots$	No greatest member
Terminal part	$\ldots, X - 2, X - 1, X$	No least member

Successor stays within the initial part. Within the terminal part it changes $X - r$ to $X - (r - 1)$, as long as $r \geq 1$. There is no last initial time followed by a first terminal time. Thus the two parts are separate *successor chains* when viewed from outside the setting.

X is a **nonstandard number**: an object of this model beyond every ordinary numeral. It is still below $X + 1$, has predecessor $X - 1$, and obeys the model's arithmetic laws. We will enlarge this setting to hold tape values and history codes while preserving exactly this interval.

5. The apparent history: genuine at the start, halting at the end

We need a second formal positive quantity Y, larger than every rational function of X. A rational function is a quotient of polynomials. To allow the tape values below, use the base ring

$$\mathcal{A} = \mathbb{Z}[X] + Y\mathbb{Q}(X)[Y] = \{p(X) + Yf(X,Y) : p \in \mathbb{Z}[X],\ f \in \mathbb{Q}(X)[Y]\}.$$

The coefficients of positive powers of Y may be rational functions of X; the constant coefficient in Y must be an integer-coefficient polynomial in X.

This includes $Y/2$, $Y/4$, $Y/(X+1)$ and $Y/2-1$, but does *not* include $1/2$. In particular, it remains discretely ordered: any positive expression involving Y exceeds all polynomials in X, and expressions without Y belong to $\mathbb{Z}[X]$. The interval $[0, X]$ is still exactly I from page 4.

Assigning a row to every time in the interval

At each ordinary time $i \in \mathbb{N}$, assign the genuine configuration of the machine. These are ordinary natural-number quadruples. All their fields are below Y^2, since each field is an ordinary finite integer and Y exceeds every such integer.

At the other times, assign the following rows, listed as (q, L, b, R). The formula with $k \geq 1$ covers all times $X-3, X-4, \ldots$.

Time	Assigned configuration
$X-2-k$, ordinary $k \geq 1$	$(0,\ Y/2^k-1,\ 1,\ 3\cdot 2^{k-1}-1)$
$X-2$	$(0,\ Y-1,\ 0,\ 1)$
$X-1$	$(1,\ 2Y-1,\ 1,\ 0)$
X	$(3,\ Y-1,\ 1,\ 1)$

Every entry is nonnegative and below Y^2. The expressions 2^k describe ordinary finite constants in our outside construction; no exponentiation function is added to the arithmetic language.

Checking the terminal transitions

Before $X-2$, the rows use $A1$ right moves. For $k \geq 2$ the necessary equalities are

$$2(Y/2^k-1)+1 = Y/2^{k-1}-1, \qquad 2(3\cdot 2^{k-2}-1)+1 = 3\cdot 2^{k-1}-1.$$

At $X-3$, the row is $(0, Y/2-1, 1, 2)$; its right move reaches $(0, Y-1, 0, 1)$ because $2 = 2\cdot 1+0$. At $X-2$, $A0$ writes 1 and enters B while reading 1. Finally $B1$ executes the encoded `1LZ` transition: $2Y-1 = 2(Y-1)+1$ and $1 = 2\cdot 0+1$, giving the last row.

Thus every adjacent pair is legal in both parts of I. The initial row is blank and the final row has state 3. What remains is essential: these assigned rows must be represented by actual history-code elements of our model.

6. Putting all those rows into one numerical code

The two-chain picture by itself is *not* a complete countermodel proof. The formula Run does not accept an externally described list. It requires four numbers B_j with the specified quotient-and-remainder equations. An infinite collection of desired remainders need not have a code in the base ring $\mathcal{A}$.

We therefore enlarge the ring, while proving that the enlargement introduces no new times below X. The construction is algebraic; it needs no compactness theorem or ultraproduct.

Names for the data

Let $v_{j,i}$ be the value assigned to column j at time $i \in I$ on page 5. For instance, $v_{0,X} = 3$, $v_{1,X} = Y - 1$, and $v_{2,X} = v_{3,X} = 1$. Every $v_{j,i}$ belongs to $\mathcal{A}$ and satisfies $0 \le v_{j,i} < Y^2$.

Use the same modulus parameter for all four columns:

$$C = Y^2, \qquad D_i = 1 + (i+1)C.$$

Introduce four independent positive symbols B_0, B_1, B_2, B_3, successively larger than every rational expression in the earlier symbols. Work temporarily in the field of rational functions

$$F = \mathbb{Q}(X)(Y)(B_0)(B_1)(B_2)(B_3).$$

The order is defined successively by the highest nonzero coefficient in each new variable; a fraction has the sign of its numerator divided by the sign of its denominator. A **field** permits division by every nonzero element. We use it as a workshop; the arithmetic domain will be a much smaller nonnegative part of a ring inside it.

For each $i \in I$ and $j = 0, 1, 2, 3$, take the element of this field

$$U_{j,i} = \frac{B_j - v_{j,i}}{D_i}.$$

Let $\mathcal{R}$ be the smallest subring of F containing $\mathcal{A}$, the four B_j, and all these $U_{j,i}$. Concretely, its elements are the results of *finite* additions, subtractions, and multiplications of these generators. There are infinitely many generators, but each individual expression uses only finitely many of them.

By the very definition of $U_{j,i}$, the desired code equations now hold:

$$B_j = U_{j,i} D_i + v_{j,i}.$$

However, adding fractions carelessly can destroy discrete order: adding $1/2$, for instance, would do so. We must prove that this construction has not done that, and has not inserted intermediate times such as $X/2$.

The crucial algebraic fact

Every finite collection of the new generators lies in an ordinary polynomial ring over $\mathcal{A}$ whose new variables are extremely large. Such variables cannot create numbers between 0 and 1, or new numbers between 0 and X. Page 7 proves this fact. It is the step that validates the coding construction rather than merely postulating a coded history.

7. Why adding the codes preserves the arithmetic

This page gives the algebra behind the preceding claim. It uses the finite Chinese remainder idea: finitely many compatible remainder requests can be interpolated. For example, the ordinary integer 23 has remainder 2 modulo 3 and remainder 3 modulo 5. We need the corresponding finite algebraic construction. [3]

Step 1: the changing divisors can be separated

For distinct $i, k \in I$, put $a = i + 1$ and $b = k + 1$, so $a - b \neq 0$. The following identity is a direct expansion:

$$\underbrace{\frac{b^2 C}{a-b}}_{\alpha}(1 + aC) + \underbrace{\left(1 - \frac{abC}{a-b}\right)}_{\gamma}(1 + bC) = 1.$$

Both coefficients belong to $\mathcal{A}$: a, b are polynomials in X, and each noninteger coefficient is multiplied by $C = Y^2$. In particular, division by $a - b$ occurs only as a coefficient of a positive power of Y, which the base ring permits.

For any ordinary finite set $J \subset I$ and each column j, there is consequently $F_j \in \mathcal{A}$ such that

$$F_j - v_{j,i} \text{ is a multiple of } D_i \text{ in } \mathcal{A} \qquad (i \in J).$$

Here is an explicit construction. For each $i \in J$, form e_i by multiplying, over $k \in J \setminus \{i\}$, the factors $\gamma_{ik} D_k$ from the identity above. Then e_i is 1 modulo D_i and 0 modulo every other D_k. Set $F_j = \sum_{i \in J} v_{j,i} e_i$. These are finite sums and products, and the stated divisibility follows.

Step 2: a finite collection fits into a polynomial ring

Take any expression defining an element of $\mathcal{R}$. Let J contain the finitely many indices of all $U_{j,i}$ used in that expression, and put $D = \prod_{i \in J} D_i$. Choose F_j as above and introduce, within the field,

$$T_j = \frac{B_j - F_j}{D}.$$

Then $B_j = F_j + D T_j$, and for every used $U_{j,i}$,

$$U_{j,i} = \frac{F_j - v_{j,i}}{D_i} + \frac{D}{D_i} T_j \ \in \ \mathcal{A}[T_0, T_1, T_2, T_3].$$

Both coefficients lie in $\mathcal{A}$. Each T_j is still successively larger than all earlier quantities, because B_j was chosen larger than every rational expression in them. There are no nonzero polynomial relations among the T_j over $\mathbb{Q}(X, Y)$: their change from the independent B_j is invertible and affine. Thus polynomial expressions in them have unambiguous coefficients.

Step 3: no new small elements appear

In a polynomial ring with a new positive variable larger than every coefficient, a positive polynomial that genuinely uses that variable exceeds every base-ring element. Apply this successively to T_3, T_2, T_1, T_0. An element of this polynomial ring between 0 and X must therefore lie in $\mathcal{A}$ already. The same argument applies between 0 and 1.

Every element of $\mathcal{R}$ belongs to one such polynomial ring. Hence $\mathcal{R}$ is discretely ordered, and its interval $[0, X]$ is exactly the original I. This argument also covers cancellations: if all new-variable terms cancel, the resulting constant coefficient is in $\mathcal{A}$. Let $\mathcal{M} = \mathcal{R}_{\geq 0}$, with inherited arithmetic and order. It is a model of PA$^-$.

8. Checking every clause of the original history formula

We now have the arithmetic setting, the assigned rows, and their numerical codes. We must still verify the bounds and the universal conditions in Run; checking one selected successor per row would be insufficient.

Choose the following elements of $\mathcal{M}$ as history parameters:

$$n = X, \qquad C_0 = C_1 = C_2 = C_3 = Y^2, \qquad H = 1 + Y^2 + B_0 + B_1 + B_2 + B_3.$$

The four B_j are the code elements from page 6.

Bounds and remainder equations

For every $i \in I$, $0 \leq v_{j,i} < C < D_i$. Also $U_{j,i} > 0$, since B_j exceeds all elements of $\mathcal{A}$, including $v_{j,i}$. From

$$B_j = U_{j,i} D_i + v_{j,i}, \qquad D_i > 1$$

it follows that $U_{j,i} < B_j < H$. Every assigned field is below $C < H$. Thus each equation gives the bounded quotient witness required by $\beta_j(i, v_{j,i})$.

Why decoding cannot give an extra, bad row

Remainders below a fixed positive divisor are unique in any discretely ordered ring. To see this, suppose $B = uD + v = u'D + v'$ with $0 \leq v, v' < D$. If $u < u'$, discreteness gives $u + 1 \leq u'$, whence

$$uD + v < (u + 1)D \leq u'D \leq u'D + v',$$

a contradiction. The reverse inequality is equally impossible. Therefore $u = u'$ and $v = v'$. This proof needs only order, discreteness, and algebra. Consequently every row decoded by $\mathsf{At}(i, c)$ is the assigned row, not merely one of several possible interpretations.

The four checks

1. **A row at every time.** In this model, $i < X + 1$ is equivalent to $0 \leq i \leq X$. Page 7 proves that these times are exactly I. At each of them the four assigned fields and quotient witnesses satisfy the required bounds.
2. **The blank start.** Time 0 belongs to the initial part, where we assigned the genuine run. Its first row is $(0, 0, 0, 0)$. Uniqueness rules out any other decoded row there.
3. **Every adjacent pair is legal.** For an ordinary i, both i and $i + 1$ lie on the genuine run. For $i = X - r < X$, with ordinary $r \geq 1$, both lie in the terminal part, whose transitions were checked on page 5. There is no third case and no adjacent pair crossing between the parts. Uniqueness supplies the check for *all* decoded tuples in the universal clause.
4. **A final halt.** The row at $n = X$ is $(3, Y - 1, 1, 1)$, so it has halting state code 3 and all its entries are below H.

We have therefore exhibited parameters satisfying the actual formula Run inside a model of PA$^-$. In symbols, $\mathcal{M} \models \mathsf{Halt}_M$.

9. Why a proof of nonhalting is impossible in PA$^-$

We can now finish the argument using soundness. [2]

1. Every PA$^-$ axiom holds in $\mathcal{M}$.
2. The fixed encoded halting sentence Halt_M is true in $\mathcal{M}$.
3. Therefore its negation $\mathsf{Nonhalt}_M$ is false in $\mathcal{M}$.
4. If PA$^-$ had a formal proof of $\mathsf{Nonhalt}_M$, soundness would make $\mathsf{Nonhalt}_M$ true in every model of PA$^-$, including $\mathcal{M}$.
5. That contradicts step 3. Hence $\boxed{\mathrm{PA}^- \nvdash \mathsf{Nonhalt}_M}$.

The ordinary natural numbers satisfy PA$^-$ and make Halt_M false, by the real nonhalting argument. They likewise rule out a PA$^-$ proof of Halt_M. Thus the fixed encoded halting sentence is **independent of** PA$^-$: the theory proves neither it nor its negation.

Does the machine somehow halt after infinitely many steps?

No. The physical computation on ordinary finite times never halts. The object X and the tape values involving Y are elements of another arithmetic setting. They are not ordinary finite integers or ordinary finite binary tapes. That setting satisfies the weak axioms and accepts the local arithmetic history checks. This is precisely why those axioms cannot rule out every encoded witness.

We have not replaced the transition equations, supplied an extra tape invariant, or changed the four history checks. The interpretation of the arithmetic domain changes. The result is consequently tied to those formulas; it is not automatically a theorem about every alternative definition of computation.

Exactly where this model fails induction

Take the quantifier-free property $\varphi(t)$ saying $2t < X$. In our model it holds exactly for ordinary $t = 0, 1, 2, \ldots$. Indeed, $2t < X$ implies $t < X$; the other possibilities $t = X - r$ give $2t > X$.

Thus $\varphi(0)$ holds, and whenever $\varphi(t)$ holds, $\varphi(t+1)$ holds. Nevertheless $\varphi(X)$ fails. This directly violates the open-induction instance

$$[\varphi(0) \wedge \forall t(\varphi(t) \to \varphi(t+1))] \to \forall t\, \varphi(t).$$

So this countermodel is *not* a model of IOpen or $I\Delta_0$. It establishes no independence from those stronger theories. Their treatment of this machine needs a separate argument.

What the lesson is, and what it is not

Nonstandardness alone does not invalidate induction: even full PA has nonstandard models. Nor do separate successor chains alone settle the question. The model built here allows a *definable* separation that violates an induction instance. Its sparse time interval, nonstandard tape values, and actual remainder codes work together to satisfy Halt_M.

This is a countermodel proof carried out in ordinary mathematics, not inside PA$^-$. A proof assistant could check such a construction in a suitable stronger mathematical foundation; the construction has not been formalized in the project's Metamath kernel.

Appendix A. The axioms and their model verification

The arithmetic language is $0, 1, +, \cdot, <, =$ with ordinary first-order logic. The PA$^-$ axioms used here are the universal closures of these formulas; in each separate formula, all displayed free variables are universally quantified.

Algebra	Order
$(x+y)+z = x+(y+z)$	$\neg(x < x)$
$x+y = y+x$	$(x < y \wedge y < z) \to x < z$
$x+0 = x$	$x < y \vee x = y \vee y < x$
$(xy)z = x(yz)$	$\neg(x < 0)$
$xy = yx$	$x < y \to x+z < y+z$
$x \cdot 1 = x$	$(x < y \wedge 0 < z) \to xz < yz$
$x \cdot 0 = 0$	$x < y \leftrightarrow \exists z\,[y = (x+z)+1]$
$x(y+z) = xy+xz$	

There is no induction schema. Ring arithmetic supplies all eight algebraic laws in the nonnegative part. Its inherited field order supplies irreflexivity, transitivity, trichotomy, compatibility with addition and positive multiplication, and the absence of negative domain elements.

For the last axiom, if $x < y$, then $d = y - x$ is a positive ring element. Discreteness gives $d \geq 1$, so $z = d - 1$ belongs to the nonnegative part and $y = (x+z)+1$. Conversely such an equality implies $x < y$, since $z \geq 0$ and $1 > 0$. This verifies every arithmetic axiom for $\mathcal{M} = \mathcal{R}_{\geq 0}$.

Appendix B. Notation and references

Notation	Meaning
$\mathbb{N}, \mathbb{Z}, \mathbb{Q}$	Ordinary naturals, integers, and rationals
$\mathbb{Q}(X)$	Rational functions of the formal symbol X
$S[T]$	Finite polynomials in T with coefficients in S
$\mathcal{R}_{\geq 0}$	Nonnegative elements of the ordered ring $\mathcal{R}$
$\mathcal{M} \models \theta$	Sentence θ is true in the interpretation $\mathcal{M}$
PA$^- \vdash \theta$	There is a formal proof of θ from PA$^-$

[1] Richard Kaye, *Models of Peano Arithmetic*, Oxford Logic Guides 15, Clarendon Press, 1991, Chapter 2, especially §2.1: discretely ordered rings and PA$^-$. Publisher's chapter page.

[2] Eric Blais, *Logic and Computation*, Lecture 15, "Soundness Theorem," University of Waterloo. `https://cs.uwaterloo.ca/~eblais/cs245e/f25/fol-soundness`.

[3] Charles Rezk, *Lecture Notes (Part 2), Math 500*, Fall 2022, section "Chinese Remainder Theorem." University-hosted notes. Page 7 gives the particular finite interpolation calculation needed here, so no infinite Chinese remainder principle is assumed.

[4] Project document *Busy Beaver certificates in bounded arithmetic* (`bb3_deciders_idelta0_paper.pdf`), §§3–4, pp. 3–5: state codes, binary stacks, and the four `Run` clauses. Those definitions are retained here with the missing transition specified as `1LZ`.

The background references do not assert this specific machine-independence theorem. The explicit ring extension and its verification are the mathematical argument presented here.